

Bachelor of Technology — Agricultural Engineering with Data Analytics

Table 1: Agricultural Engineering with Data Analytics — B26 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	36
	IC Basket	6
	IKS + HSS	15
Discipline Courses	Discipline Core	49
	Discipline Electives	17
Electives	Free Electives	25
	MTP (4+4, equivalent as DE)	8
	ISTP (equivalent as FE)	4
Total		160

A dash (—) in the Course Code column means that the source does not yet notify an official catalogue code.

Table 2: B. Tech Agricultural Engineering with Data Analytics — B26 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CE-310	Strength of Materials and Structures	3-0-0-3	3
2	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
3	—	Crop Production Technology		3
4	—	Soil Science and Mechanics		4
5	—	Farm Machinery and Equipment		3
6	CE-201	Surveying: Traditional and Digital	2-0-4-4	4
7	CE-251	Hydraulics Engineering	3-0-0-3	3
8	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
9	EE-302	Control Systems	3-0-2-4	4
10	—	Theory and Design of Machine Elements		4
11	CE-303	Water Resources Engineering	3-0-0-3	3
12	CE-353P	Civil Engineering Drawing	0-0-2-1	1
13	CE-451	Irrigation Engineering and Hydraulic Structures	3-0-0-3	3
14	CS-541P	IoT Systems and Clouds	1-0-3-3	3
15	—	Food Science and Technology and Lab		4
16	—	Self-Sufficient Farming (Natural Farming)		2
17	—	Post-Harvest Engineering		3
Total				49

Table 3: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-112	Calculus	1.5-0.5-0-2	2
2	IC	IC-114	Linear Algebra	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-230	Environmental Science	3-0-0-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		IC-182	History of Science and Technology	3	
8	IKS + HSS (choose 1)	HS-108	Basic English for Engineers	3-0-0-3	3
9		HS-112	Japanese Language I	3-0-0-3	
10		HS-342	German I	3-0-0-3	
11		IK-101	Sanskrit language-Level 1		
Total					21

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
3	IC	IC-115	ODE and Integral Transforms	2-0-0-2	2
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-240	Rigid Body Dynamics	1.5-1.5-0-3	3
Total					22

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-202P	Design Practicum	0-0-6-3	3
2	IC	IC-272	Machine Learning	2-0-2-3	3
3	DC	CE-310	Strength of Materials and Structures	3-0-0-3	3
4	DC	CE-310P	Strength of Materials and Structures Lab	0-0-2-1	1
5	DC	—	Crop Production Technology		3
6	DC	—	Soil Science and Mechanics		4
7	DC	—	Farm Machinery and Equipment		3
Total					20

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-222P	Physics Practicum	0-0-3-2	2
2	DC	CE-201	Surveying: Traditional and Digital	2-0-4-4	4
3	DC	CE-251	Hydraulics Engineering	3-0-0-3	3
4	DC	CE-304P	Hydraulics Engineering Lab	0-0-2-1	1
5	DC	EE-302	Control Systems	3-0-2-4	4
6	DC	—	Theory and Design of Machine Elements		4
Total					18

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CE-303	Water Resources Engineering	3-0-0-3	3
2	DC	CE-353P	Civil Engineering Drawing	0-0-2-1	1
3	DC	CE-451	Irrigation Engineering and Hydraulic Structures	3-0-0-3	3
4	DC	CS-541P	IoT Systems and Clouds	1-0-3-3	3
5	DC	—	Food Science and Technology and Lab		4
6	DC	—	Self-Sufficient Farming (Natural Farming)		2
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	—	Post-Harvest Engineering		3
2	ISTP	DP-301P	ISTP or Equivalent	0-0-6-4	4
Total					7

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	MTP	DP-401P	MTP-1	0-0-6-4	4
Total					4

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. Tech Agricultural Engineering with Data Analytics — B26 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-010	Internship		2
2	MTP	DP-402P	MTP-2	0-0-6-4	4
Total					6

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 17 credits of **Discipline Electives** and 25 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

The following compulsory project components count toward the DE credit requirement; they are not selectable DE options.

Table 11: Agricultural Engineering with Data Analytics — Compulsory project components credited as DE

Sl. No	Course Code	Course Name	Credits
1	DP-401P	MTP-1 (compulsory; counted as DE)	4
2	DP-402P	MTP-2 (compulsory; counted as DE)	4